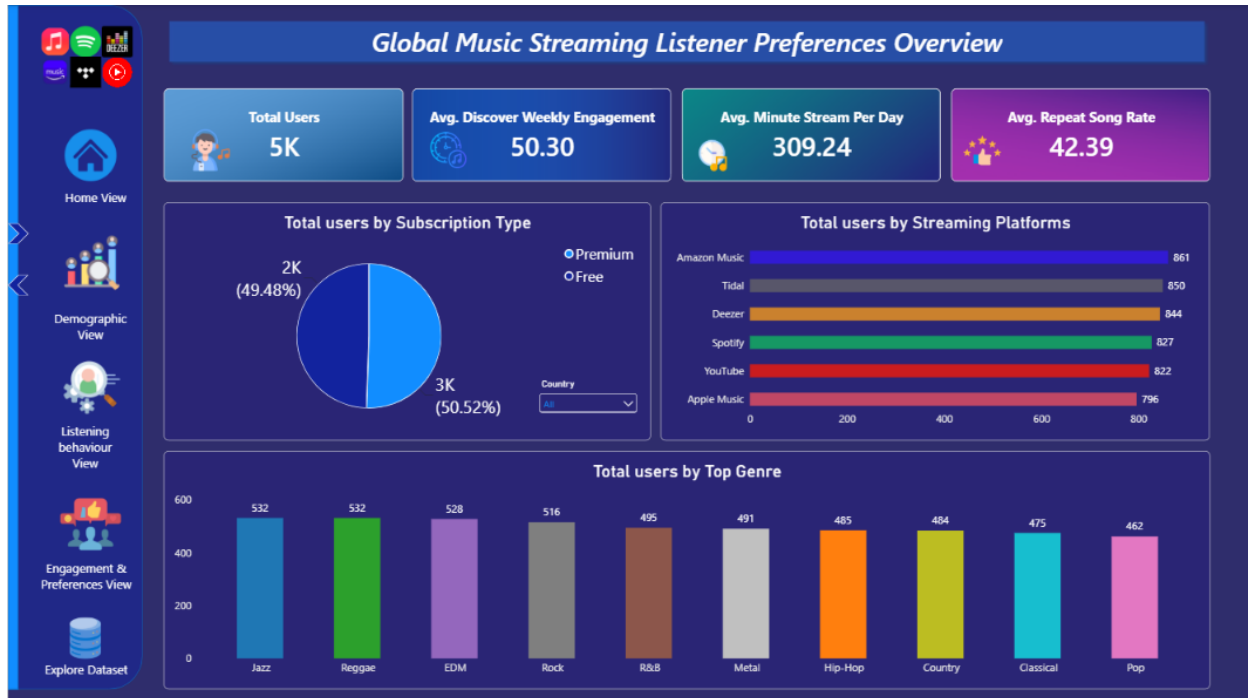


Dashboard Analysis

A. Overview Dashboard

This section provides a high-level summary of music streaming trends.



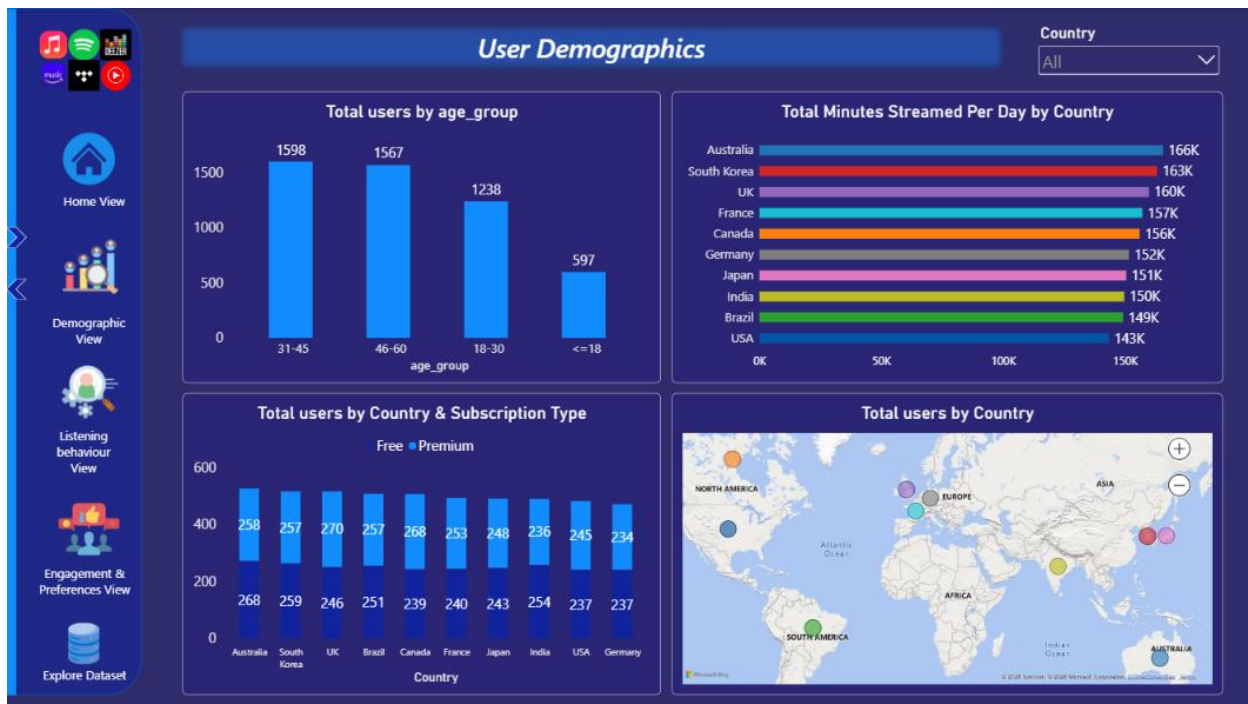
- **Total Users:** 5K users are analyzed in the dataset.
- **Average Discover Weekly Engagement :** 50.30 % users have weekly engagement
- **Average Minutes Streamed per Day :** About 309 minutes are streamed daily
- **Average Repeat Song Rate :** About 40% is the repeat song rate.
- **Subscription Type:**
 - Free Users: 49.48% (Approx. 2K users)
 - Premium Users: 50.52% (Approx. 3K users)
- **Top Streaming Platforms:**
 - Amazon Music, Tidal, Deezer, Spotify, YouTube, and Apple Music have significant user bases.
- **Popular Genres:**

- Jazz, Reggae, EDM, Rock, R&B, Metal, Hip-Hop, Country, Classical, and Pop are the most played genres.

Key Insight: Users are almost evenly split between free and premium subscriptions. Major streaming platforms have a competitive market share.

B. User Demographics

This section explores the characteristics of users based on age and geographic location.



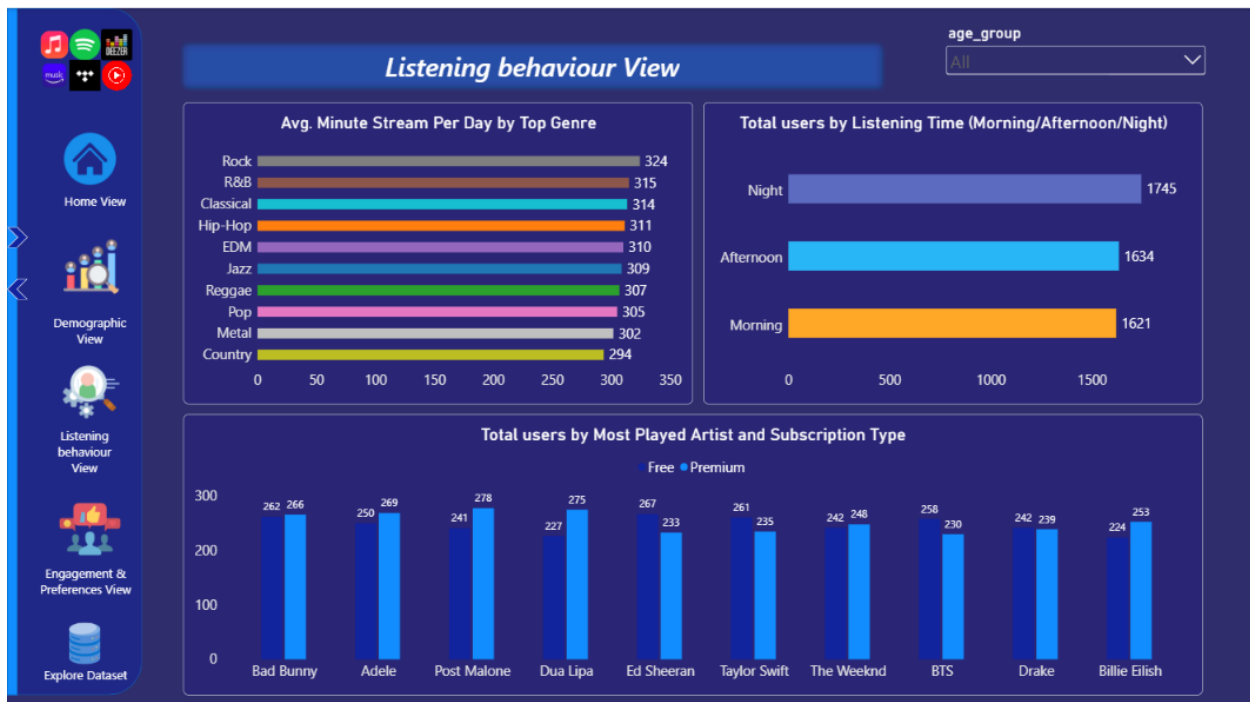
- **User Age Groups:**
 - The largest user groups belong to the 31-45 and 46-60 age brackets.
 - A notable number of users are in the 16-30 age range.
 - Very few users fall under the 0-15 category.
- **Minutes Streamed Per Day by Country:**
 - Countries like Australia, South Korea, UK, France, Canada, Germany, Japan, and India show similar streaming patterns, with an average daily streaming time between 143K to 166K minutes.
- **Users by Country and Subscription Type:**

- The premium-to-free user ratio is consistent across different countries.

Key Insight: The 31-60 age group dominates music streaming, making them a key target audience.

C. Listening Behavior of Users

This section highlights user habits related to music consumption.



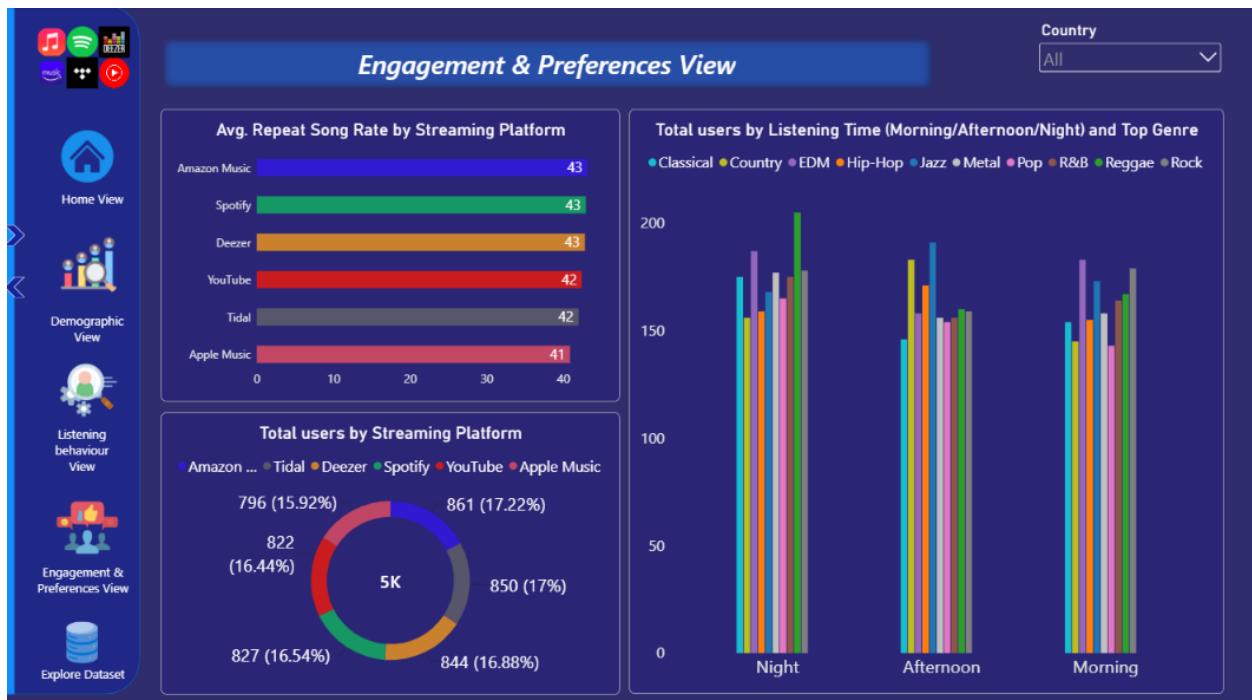
- **Average Time Spent Per Day by Genre:**
 - Users spend the most time listening to Rock, R&B, and Classical music, followed by Hip-Hop, EDM, and Jazz.
- **Most Played Artists by Subscription Type:**
 - Popular artists include Bad Bunny, Dua Lipa, The Weekend, BTS, Ed Sheeran, Adele, Post Malone, and Taylor Swift.
 - There is a balanced mix of premium and free users among different artist fan bases.
- **Listening Time Distribution:**

- Night time has the highest number of users, followed by Afternoon and Morning sessions.

Key Insight: Rock and R&B are the most engaging genres. Users mostly stream music at night.

D. Engagement & Preferences

This section examines song repeat rates, platform engagement, and genre-based listening time preferences.



User_ID	Age	CountryFlag	Country	Streaming Platform	Top Genre	Minutes Streamed Per Day	Number of Songs Liked	Most Played Artist	Subscription Type	Liste
U1000	34		Japan	Tidal	Reggae	295	138	Adele	Free	
U1001	24		Germany	Deezer	Country	86	388	Ed Sheeran	Premium	
U1002	49		Germany	Deezer	Pop	363	368	Post Malone	Premium	
U1003	55		Australia	YouTube	Reggae	348	349	Dua Lipa	Premium	
U1004	13		Germany	Amazon Music	Reggae	30	328	Adele	Free	
U1005	25		South Korea	Spotify	EDM	536	243	The Weeknd	Premium	
U1006	49		Australia	Tidal	Country	547	70	Billie Eilish	Free	
U1007	32		UK	Amazon Music	Classical	98	448	Dua Lipa	Premium	
U1008	25		Germany	Tidal	Metal	526	27	BTS	Premium	
U1009	37		Brazil	Spotify	Reggae	514	299	Billie Eilish	Premium	
U1010	40		South Korea	Deezer	Reggae	266	462	Taylor Swift	Premium	
U1011	43		Brazil	Spotify	Rock	351	108	Drake	Free	
U1012	46		Canada	Deezer	Rock	389	328	Ed Sheeran	Premium	
U1013	57		Germany	Tidal	Rock	506	432	Drake	Free	
U1014	23		India	Apple Music	Hip-Hop	502	215	Dua Lipa	Premium	
U1015	33		UK	Spotify	Hip-Hop	193	11	Drake	Premium	
U1016	19		India	Spotify	Country	542	499	Adele	Free	

- **Repeat Song Rate by Streaming Platform:**
 - Users on Amazon Music and Spotify have the highest repeat song rates.
 - Deezer and YouTube users show moderate repeat behavior.
- **User Count by Streaming Platform:**
 - The user distribution across platforms is almost balanced, but Amazon Music is a little dominating among all the other platforms.
- **Listening Time Preference by Genre:**
 - Certain genres are more preferred at specific times of the day.
 - For example, Reggae and Classical music are streamed more at night, while Jazz and Country are popular in the afternoon and Rock in the morning.

Key Insight: High repeat rates on platforms like Amazon Music and Spotify indicate strong user loyalty.

Business Questions and Answers:

I. What are the most popular music streaming platforms?

Ans:

Most Popular Music Streaming Platforms

- Amazon Music, Spotify, YouTube, Apple Music, Deezer, and Tidal all have significant user bases.
- Amazon Music shows slightly higher dominance compared to others.
- **Insight:** The market is competitive, with no single platform overwhelmingly leading.

II. How does age impact music preferences?

Ans:

Impact of Age on Music Preferences

- The 31–45 and 46–60 age groups are the largest user segments.
- The 16–30 group is also notable, while 0–15 is minimal.
- **Insight:** Middle-aged users (31–60) dominate streaming, making them the prime target audience for marketing and personalised features.

III. What are the most streamed genres and artists?

Ans:

Most Streamed Genres and Artists

- **Genres:** Rock, R&B, and Classical are the most engaging, followed by Hip-Hop, EDM, and Jazz.
- **Artists:** Bad Bunny, Dua Lipa, The Weeknd, BTS, Ed Sheeran, Adele, Post Malone, and Taylor Swift are among the most played.
- **Insight:** Rock and R&B drive the highest listening time, while global pop stars attract both free and premium users.

IV. How do free vs. premium users differ in streaming behaviour?

Ans:

Free vs. Premium User Behaviour

- **Split:** Nearly equal — Free (49.48%) vs Premium (50.52%).
- **Behaviour:** Premium users show higher engagement, especially with top artists.
- **Insight:** Free users can be converted through exclusive features, personalised playlists, and targeted promotions.

V. What time of day do users stream music the most?

Ans:

Time of Day for Streaming

- Nighttime is the peak streaming period, followed by Afternoon and Morning.
- **Genre Preferences by Time:**
 - Rock → Morning
 - Jazz & Country → Afternoon
 - Reggae & Classical → Night
- **Insight:** Platforms should push new releases and promotions during nighttime for maximum impact.

VI. *Are there any regional trends in music streaming preferences?*

Ans:

Regional Trends in Streaming Preferences

- Countries like Australia, South Korea, UK, France, Canada, Germany, Japan, and India show similar daily streaming patterns (143K–166K minutes).
- **Regional Variations:**
 - UK & Brazil → Nighttime preference
 - Japan & South Korea → Morning/Afternoon preference
- **Insight:** Localised content and region-specific campaigns can boost engagement and monetisation.

Key Insights and Recommendations

- **Subscription Balance:**
 - Since premium and free users are nearly equal, platforms can introduce exclusive premium features to attract more paying subscribers.
- **Genre-Based Targeting:**
 - Rock and R&B listeners spend the most time streaming music.
 - Streaming services can design personalized playlists and recommendations based on this behavior.
- **Listening Time Optimization:**
 - Users prefer streaming at night. Platforms should push new releases and promotions during peak hours.
- **Regional Expansion:**
 - Since streaming is evenly distributed across multiple countries, localized content and region-specific marketing can help improve user engagement.

Conclusion

The project revealed valuable insights into global music streaming behavior across countries and user types. Premium users consistently show higher engagement, especially for top artists like Dua Lipa and Post Malone. Listening time varies by region, with nighttime being most popular in countries like the UK and Brazil, while morning and afternoon preferences stand out in Japan and South Korea. Repeat song rates are highest for artists like Adele, Billie Eilish, and Ed Sheeran, showing strong fan loyalty. These patterns can help tailor content release timings, promotional campaigns, and artist engagement strategies. Free users can be targeted with personalized conversion offers. Country-specific preferences support localized marketing and tour planning. Overall, data-driven strategies can enhance user experience and platform growth.

Business Recommendations:

- Streaming platforms should enhance personalized recommendations to improve engagement.
- Increase exclusive content and premium features to convert free users into paid subscribers.
- Optimize marketing campaigns based on age-wise and region-wise preferences.

- Use AI-driven playlist curation to improve Discover Weekly Engagement.
- Streaming platforms should invest in AI-driven recommendations, exclusive content, and user-specific promotions to retain and grow their user base.
- Identify regions where free users have high engagement and roll out special promotions or partnerships to improve monetization without heavy price incentives.
- Send push notifications or in-app messages for new releases during preferred listening hours per country, improving click-through and listening rates.